★★★★★ **Fang-tastic reviews** rated 4.8 out of 5! **Flies tonight** order in the next 4h 32m **Worldwide delivery** tracked, duties calculated at checkout **No fear returns** we're here to help

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/ Inky Impression 5.7" (7 colour ePaper/eInk HAT)

★★★★★ 37 reviews Add to wishlist

"As for me, I am tormented with an everlasting itch for things remote. I love to sail forbidden seas, and land on barbarous coasts."

We no longer stock this product

It's sad to say goodbye but sometimes we have to retire products to make way for new things.

Need a large quantity or a customised version? We may be able to help! Contact wholesale@pimoroni.com to discuss a custom build.

> ★★★★☆
> Again a very nice epaper display, an upgrade to the wHat. But I still don't understand why Pimoroni only provides a single GPIO on the connector. Why?? Pimoroni should know that you never can have enough GPIOs ...

Bold, will hold! A big, beautiful, 5.7" (600 x 448 pixel) 7 colour electronic paper display for Raspberry Pi.

This display is **considerably bigger** than Inky wHAT (check out the comparison photo!) with plenty of screen real estate for text or graphics. The low power consumption e-paper display is **crisp and readable** in bright sunlight and the image will persist when unpowered. In a first for the Inky series, we've also added **four tactile buttons** so you can control what's on the screen without the need for extra hardware. The buttons are side-press, so they're accessible even if the display is mounted on a wall.

But the best bit is that this time you get **seven whole colours** to play with, which means this Inky is very suitable for displaying graphics, drawings or art - we've found pixel art, panels from comics and retro video game art look particularly good.

We can think of tons of uses for this lovely display - its size makes it ideal for displaying detailed graphs and charts from weather stations and sensors. You could build your own e-ink reader, or set it up as a digital dashboard for your kitchen with calendar info and reminders. It would also make a nice rotating display for showing off children's art (or your own art!).

Please note that Inky Impression takes around 30 seconds to refresh the screen, so is best suited to projects that don't rely on constant screen updates.

Features

- 5.7" EPD display (600 x 448 pixels)
  - E Ink Gallery Palette™ 4000 ePaper
  - ACeP (Advanced Color ePaper) 7-color with black, white, red, green, blue, yellow, orange.
  - Ultra wide viewing angle – >170°
  - Dot pitch – 0.1915 x 0.1915mm
- 40-pin extension header included to boost height for full-size Pis
- Standoffs included to securely attach to your Pi
- Additional pins, including I2C and SPI, broken out
- Compatible with all 40-pin header Raspberry Pi computers*
- Python library
- Comes fully assembled

Multi-colour EPD displays, like the one on Inky Impression, use ingenious electrophoresis to pull coloured particles up and down on the display. The coloured particles reflect light, unlike most display types, meaning that they're visible under bright lights. It takes approximately 30 seconds to refresh the display.

Everything comes fully-assembled, and there's no soldering required (as long as your Raspberry Pi has a 40 pin header attached). The display is securely stuck down to the Inky Impression PCB and connected via a ribbon cable. Just pop Inky Impression on your Pi and run our installer to get everything set up!

We've also broken out some handy pins including I2C and SPI, on the back of Inky Impression, letting you connect additional devices like our breakouts and allowing you to show their data right on the display.

Inky Impression will work with any version of the Pi that has a 40 pin header, including Zero variants. If you want to use it with a Raspberry Pi 400, you'll probably also want to pick up a GPIO extender cable (unless you're into using screens at a highly unusual angle).

Software

Our Python library takes the stress out of displaying text and images on Inky Impression, and we've put together a few new examples to show off Inky Impression's capabilities.

Notes

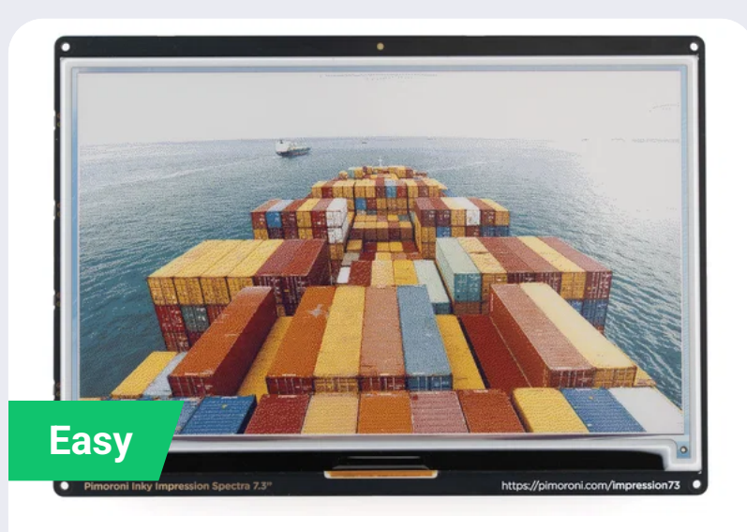
- **The Inky Impression display is made from glass so it's pretty fragile. Be careful not to drop it or press too hard on it, or it will crack. When fitting it to your Pi, grip at the edges of the board rather than pressing on top of the screen.**
- Take care when using your own fixings! It is possible to damage the display if you use standoffs/bolts with longer screw threads than those supplied in the kit.
- We got best results from this display when refreshing it at an ambient room temperature, if the screen is cold you might find that the colours are less vibrant. Solidarity to our chilly production team who discovered this!
- Overall dimensions: 125.4 x 99.5 x 9mm (W x H x D, depth includes header and display)
- Display usable area dimensions: 114.9 x 85.8mm (W x H)
- Licensed Image: Bobbi Freelance / Pixels.com

Resources

- Pinout
- Schematic
- Dimensional drawing

Guides & tutorials

Displaying Comics on Inky Impression
Learn how to download and display comic art on Inky Impression.



Getting Started with Inky Impression
Learn how to install and get started with Inky Impression.

Reviews

★★★★★ rated 4.86 out of 5 from 37 reviews

★★★★★ last year

Great quality screen, perfect for my weather station project. Quick delivery too.

Amandeep from Hackney

★★★★★ 3 years ago

amazing display, it looks so different than LCD or LED displays

Sergeï

★★★★★ 3 years ago

A customer

★★★★★ 4 years ago

This eInk display is gorgeous and the tool supplied to program it using Python3 is phenomenal. With very little python experience I was able to get it up and running quite quickly (setting up the Raspberry PI was harder) and use it as an electronic picture frame.

After about a year the display began to exhibit vertical streaks even after cleaning. I contacted Pimoroni technical support and they immediately sent me a replacement display saying it was a non-repairable fault. Amazing customer service and very satisfied.

Joshua

★★★★★ 5 years ago

A beautiful little display, the eInk looks gorgeous. A Pi Zero (when I finally got one) and some shonky Python and it now gives everyone the current weather conditions from our NetAtmo, the weather forecast, an overview of the family calendars and a little motivational boost (apparently my hair looks fab today) at a glance.

The buttons are great too. I've already set one up to pull a random xkcd and am trying to think of nifty things for the others.

As noted elsewhere, the colour can look a little washed out but turning up the saturation sorts that out, and the refresh is slow (fun to watch at first, but the novelty doesn't last long!) so it's no good for projects that need a regular refresh.

James

★★★★★ 2 years ago

I bought it from Japan, but it arrived in a very careful package.

I was impressed that it could display seven colors on electronic paper, so I bought it, and it lived up to my expectations. It is very colorful and has a retro photo-like impression, which I am enjoying.

Recommended for IoT such as photo frames and infrequently updated calendars!

It is easy to connect to the Raspberry pi, and you can get it working right away.
However, if you use screws other than those included to secure it to the spacer, be careful of the length. This is because it will shatter and destroy the glass display (I'm telling you, but I've experienced it...)

A customer

★★★★★ 4 years ago

Palermo from Toronto

★★★★☆ 4 years ago

Quite an impressive little display, though the saturation is less than you might think - if you imagine everything's going to be painted by Georges Seurat you're pretty much there.
Pretty simple to use with the python library, and it's very easy to fit.

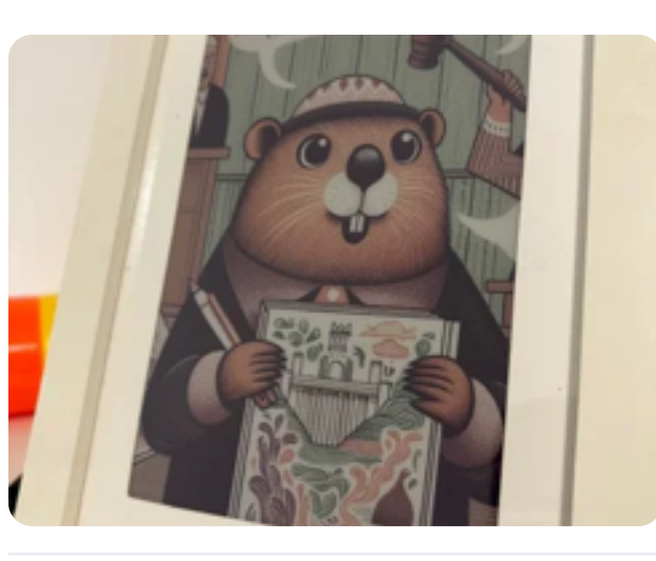
Knocked a star off because of the lack of saturation, updates make the entire screen flash many different colours, and the fact it takes ~40-45 seconds to update from an rPi0. It's completely fine for geek in-home usage, but it lacks a bit of polish if you're looking for a one-stop info screen.

What I'd really like to see on the next version is (much quicker) partial screen updates, a quicker overall update time and slightly bolder colours.

A customer

★★★★★ 3 years ago

Awesome product with superb quality. I really enjoyed building something creative with the ePaper HAT. And since the hardware as well es the open source library are basically plug and play, I could focus on the creative part. 10/10 fully recommend this HAT.



Julian from United States

★★★★★ 4 years ago

I must say I'm very happy how my images appear on this 7 colour display. Even if I don't dither the images beforehand. I've used my Inky Impression to display my cycling statistics from Strava and I'm really happy how this turned out. It would be great if there would ever be a larger version of this. I could do all kinds of things with it.

Marcel-Jan

★★★★★ 5 years ago

Love this display! Beautiful e-paper in 7 colors. Easily readable in bright conditions. Convenient HAT format plugs right onto my raspberry pi.

Mike from Glendora

★★★★★ 5 years ago

It looks way better than the 600x448 pixels, good documentation, I can only recommend this display!

Rune

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The Good Ship Pimoroni

We're here to help curious people make interesting things. Robots, gadgets, home automation, blinking lights, questionable contraptions - whatever you're building, we reckon technology is much more fun when you can poke at it, pull it apart and make it your own.

We design kits, breakout boards and all manner of colourful electronic oddities to make getting started easy and experimenting fun - from your first blinking LED to your latest gloriously overcomplicated creation.

More about our story

Useful links (we hope)

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